

Name: _____

Class: _____

SYDNEY TECHNICAL HIGH SCHOOL

YEAR 12

HSC ASSESSMENT TASK 3

JUNE 2008

MATHEMATICS Extension 1

Time Allowed: 70 minutes

Instructions:

- Write your name and class at the top of each page
- All necessary working must be shown. Marks may be deducted for careless or badly arranged work.
- Marks indicated are a guide only and may be varied if necessary.
- Start each question on a new page.
- Standard integrals can be found on the last page.

Question 1	Question 2	Question 3	Question 4	Question 5	Total
/12	/12	/12	/12	/12	/60

QUESTION1 - 12 Marks

- a) Find i) $\int \frac{x}{9+x^2} dx$ 2
- ii) $\int \frac{1}{9+x^2} dx$ 2
- b) Solve for x
- $\log_2 x = \log_2 10 - \log_2(x-3)$ 3
- c) Differentiate i) 5^x 1
- ii) $x^2 \sin^{-1} 2x$ 2
- d) Find the exact value of $\tan \left(\cos^{-1} \left(\frac{-3}{4} \right) \right)$ 2

QUESTION 2 (Start a new page) - 12 Marks

- a) Solve $\cos^2 \theta - \sin^2 \theta = 0.1$ for $0 \leq \theta \leq \pi$ 3
(answer(s) in radians correct to 2 decimal places)
- b) Find the general solution for $\sin \theta = \frac{1}{\sqrt{2}}$ 2
- c) i) Write $x^2 + 6x + 10$ in the form $(x+a)^2 + b$ 1
- ii) Hence find $\int \frac{dx}{x^2 + 6x + 10}$ 2
- d) i) Sketch $y = \sin^{-1} x$ 1
- ii) Find the exact area bounded by
 $y = \sin^{-1} x$, the x axis and the line $x = 1$ 3

QUESTION 3 (Start a new page) - 12 Marks

- a) i) Find $\frac{d}{dx} \sqrt{1-x^2}$ 1
- ii) Using part i) show that
- $$\int_0^1 \frac{1+x}{\sqrt{1-x^2}} dx = \frac{\pi}{2} + 1$$
- 3
- b) i) If $f(x) = (x-1)^2$ for $x \leq 1$, find $f^{-1}(x)$ and state its domain and range 3
- ii) Find any points(s) of intersection of $y = f(x)$ and $y = f^{-1}(x)$ 2
- c) i) If $\tan^{-1} x = \alpha$ and $\tan^{-1} y = \beta$ prove that
- $$\tan^{-1} x + \tan^{-1} y = \tan^{-1} \left(\frac{x+y}{1-xy} \right)$$
- 2
- ii) Hence evaluate $\tan^{-1} \left(\frac{1}{2} \right) + \tan^{-1} \left(\frac{1}{3} \right)$ (in exact form) 1

QUESTION 4 (Start a new page) - 12 Marks

- a) Find $\int \frac{dx}{\sqrt{x}(1+x)}$ using the substitution $u = \sqrt{x}$ or otherwise 3
- b) A cylindrical solid of height 10cm is being turned on a cutting machine so that the radius is being reduced by 0.3cm/min.
- Find at what rate the surface area is decreasing, when the radius is 5cm (in exact form)
- (surface area = $2\pi r^2 + 2\pi rh$) 3

- c) The rate of cooling of an object is proportional to the excess of the object's temperature above the surrounding temperature, $\frac{dT}{dt} = k (T - T_0)$

T is the object's temperature

T_0 is the surrounding temperature.

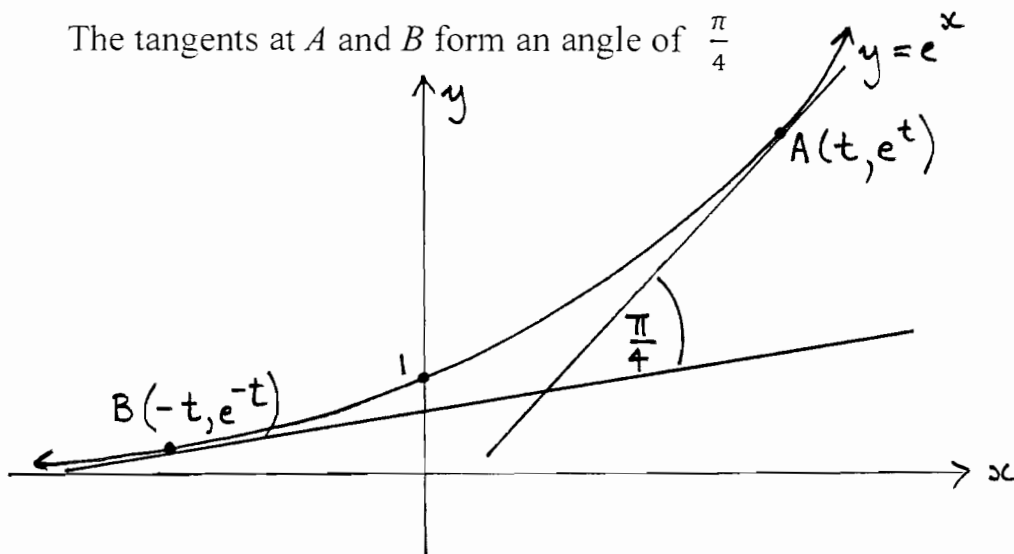
A pot of hot water cools from 90°C to 85°C in 1 minute at a room temperature of 30°C .

- i) Show that $T = T_0 + Ae^{kt}$ satisfies the above equation 1
- ii) Find the exact values of A and k . 2
- iii) How long would it take to cool to 60°C ? (nearest second) 2
- iv) What would be the temperature after 4 minutes? (2 dec. places) 1

QUESTION 5 (Start a new page) **12 Marks**

- a) $A(t, e^t)$ and $B(-t, e^{-t})$ are points on the curve $y = e^x$, where $t > 0$.

The tangents at A and B form an angle of $\frac{\pi}{4}$



- i) Prove that $e^t - e^{-t} = 2$ 2
- ii) Solve this equation to prove $t = \ln(\sqrt{2} + 1)$ 2

- b) Find $\int \sin^2 3x \, dx$ 2
- c) P is the point of intersection of the graphs $y = \tan x$ and $y = A \sin x$ where $A > 1$. The x co-ordinate of P is α , and α lies between 0 and $\frac{\pi}{2}$
- i) Sketch $y = \tan x$ and $y = A \sin x$ on the same axes 2
for $0 \leq x \leq \frac{\pi}{2}$ Label the point P
- ii) Prove $\cos \alpha = \frac{1}{A}$ at P 1
- iii) If 0 is the origin, prove that the area enclosed by the arcs OP ,
on both graph is $(A - 1 - \ln A)$ unit² 3

(End of Paper)

STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

NOTE : $\ln x = \log_e x$, $x > 0$

Question 1

a) i) $\int \frac{x}{q+x^2} dx = \frac{1}{2} \int \frac{2x}{q+x^2} dx$
 $= \frac{1}{2} \ln(q+x^2) + c$

ii) $\int \frac{1}{q+x^2} dx = \frac{1}{\sqrt{q}} \tan^{-1} \frac{x}{\sqrt{q}} + c$

b) $\log_2 x = \log_2 10 - \log_2 (x-3)$

$\log_2 x = \log_2 \left(\frac{10}{x-3} \right)$
 $x = \frac{10}{x-3}$

$x^2 - 3x = 10$

$x^2 - 3x - 10 = 0$

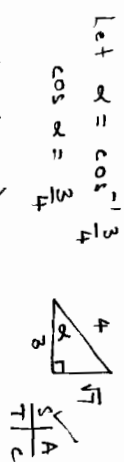
$(x-5)(x+2) = 0 \therefore x = 5 \text{ only}$

c) i) $\frac{d}{dx} (5^x) = \ln 5 \cdot 5^x$

ii) $u = x^2 \quad v = \sin^{-1} 2x$
 $u' = 2x \quad v' = \frac{2}{\sqrt{1-4x^2}}$

$\therefore \frac{d}{dx} (x^2 \sin^{-1} 2x) = 2x \sin^{-1} 2x + \frac{2x^2}{\sqrt{1-4x^2}}$

d) $\tan(\cos^{-1}(-\frac{3}{4})) = \tan(\pi - \cos^{-1} \frac{3}{4})$



$\therefore \tan(\pi - \alpha) = -\tan \alpha$
 $= -\frac{4}{3}$

Question 2

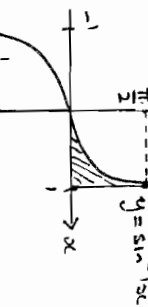
a) $\cos^2 \theta - \sin^2 \theta = 0.1$
 $\cos 2\theta = 0.1$
 $2\theta = 1.4706, 4.8126$
 $\theta = 0.735, 2.406$

b) $\sin \theta = \frac{1}{\sqrt{2}}$

$\theta = n\pi + (-1)^n \sin^{-1}(\frac{1}{\sqrt{2}})$
 $\theta = n\pi + (-1)^n \cdot \frac{\pi}{4}$
 where n is an integer

c) i) $x^2 + 6x + 10 = (x^2 + 6x + 9) + 1$
 $= (x+3)^2 + 1$

ii) $\int \frac{dx}{(x+3)^2 + 1} = \tan^{-1}(x+3) + c$



$A = \int_0^1 \sin^{-1} x dx$

$= \frac{\pi}{2} - \int_0^1 \sin y dy$
 $= \frac{\pi}{2} - [-\cos y]_0^{\pi/2}$
 $= \frac{\pi}{2} + [0 - 1]$
 $= (\frac{\pi}{2} - 1) \text{ unit}^2$

Question 3

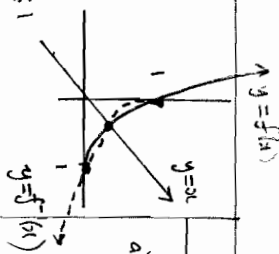
a) i) $\frac{d}{dx} (1-x^2)^{1/2} = \frac{1}{2} x^{-1/2} (1-x^2)^{-1/2}$

ii) $\int \frac{1+x}{\sqrt{1-x^2}} dx = \int \frac{1}{\sqrt{1-x^2}} + \frac{x}{\sqrt{1-x^2}} dx$

$= \left[\sin^{-1} x - \sqrt{1-x^2} \right]_0^1$
 $= \frac{\pi}{2} - (0 - 1)$
 $= \frac{\pi}{2} + 1$

b) i) $f(x) = (x-1)^2$

$x = (y-1)^2$
 $\pm \sqrt{x} + 1 = y$
 $f^{-1}(x) = -\sqrt{x} + 1$



ii) pt. int $x = (x-1)^2$
 $x^2 - 3x + 1 = 0$

$x = \frac{3 \pm \sqrt{9-4}}{2}$

$x = \frac{3 \pm \sqrt{5}}{2}$ from above
 D and R

at $x = \frac{3-\sqrt{5}}{2}$

c) i) $\tan^{-1} x = \alpha \quad \tan^{-1} y = \beta$

$\therefore x = \tan \alpha \quad y = \tan \beta$
 $\tan(\alpha + \beta) = \frac{\tan \alpha + \tan \beta}{1 - \tan \alpha \tan \beta}$

$\tan(\alpha + \beta) = \frac{x+y}{1-xy}$

$\alpha + \beta = \tan^{-1} \left(\frac{x+y}{1-xy} \right)$

$\tan^{-1} x + \tan^{-1} y = \tan^{-1} \left(\frac{x+y}{1-xy} \right)$

ii) $\tan^{-1}(\frac{1}{2}) + \tan^{-1}(\frac{1}{3}) = \tan^{-1} \left(\frac{\frac{1}{2} + \frac{1}{3}}{1 - \frac{1}{2} \cdot \frac{1}{3}} \right)$
 $= \tan^{-1}(1)$

$\tan^{-1}(\frac{1}{2}) + \tan^{-1}(\frac{1}{3}) = \frac{\pi}{4}$

Question 4

a) $u = \sqrt{x} = x^{1/2}$
 $\frac{du}{dx} = \frac{1}{2} x^{-1/2} = \frac{1}{2\sqrt{x}}$

$du = \frac{1}{2\sqrt{x}} dx$

$\therefore dx = du \cdot 2\sqrt{x}$

$\int \frac{dx}{\sqrt{x}(1+x)} = \int \frac{du \cdot 2\sqrt{x}}{\sqrt{x}(1+u^2)}$

$= 2 \int \frac{du}{1+u^2}$

$= 2 \tan^{-1} u + c$
 $= 2 \tan^{-1} \sqrt{x} + c$

b) $A = 2\pi R^2 + 2\pi R \cdot 10$

$\frac{dA}{dR} = -3 \text{ cm/min}$

$\frac{dA}{dR} = 4\pi R + 20\pi$

$\frac{dA}{dt} = \frac{dA}{dR} \cdot \frac{dR}{dt}$

$= (4\pi R + 20\pi) \cdot (-3)$
 sub $R = 4$

$\frac{dA}{dt} = -12\pi \text{ cm}^2/\text{min}$

c) i) $T = T_0 + Ae^{kt}$

since $T - T_0 = Ae^{kt}$

$\therefore \frac{dT}{dt} = k(T - T_0)$

ii) $T = 30 + Ae^{kt}$
 $90 = 30 + A$
 $\therefore A = 60$

$$A = 60 \quad T = 85 \quad t = 1$$

$$S = 30 + 60e^k$$

$$55 = e^{tk}$$

$$60$$

$$\left(\frac{11}{12}\right) = e^k$$

$$= 60^\circ \text{ find } t$$

$$= 30 + 60e^{\ln(\frac{11}{12})t}$$

$$0 = 60e^{\ln(\frac{11}{12})t}$$

$$\frac{1}{2} = \ln\left(\frac{11}{12}\right)t$$

$$t = \frac{\ln(\frac{1}{2})}{\ln(\frac{11}{12})}$$

$$t = 7.97 \text{ min}$$

$$t = 7 \text{ min } 58 \text{ sec}$$

$$= 4 \text{ find } T$$

$$= 30 + 60e^{\ln(\frac{11}{12}) \cdot 4}$$

$$T = 72.36^\circ \text{C}$$

$$\text{tion 5}$$

$$y = e^x \quad \frac{dy}{dx} = e^x$$

$$= \text{ at } A$$

$$-t \text{ at } B$$

$$\Pi = \left| \frac{e^{-t}}{1 + e^{-t}} \right|$$

$$1 = \frac{e^{-t}}{1 + 1}$$

$$t = -t$$

$$e^{-t} = 2$$

$$\text{ii) Let } u = e^t$$

$$u - u^{-1} = 2$$

$$u - \frac{1}{u} = 2$$

$$u^2 - 1 = 2u$$

$$u^2 - 2u - 1 = 0$$

$$u = \frac{2 \pm \sqrt{4 - 4(-1)}}{2}$$

$$u = \frac{2 \pm \sqrt{8}}{2} = 2(\pm\sqrt{2})$$

$$\therefore u = 1 \pm \sqrt{2}$$

$$\therefore e^t = 1 + \sqrt{2} \quad \text{or} \quad e^t = 1 - \sqrt{2}$$

$$\ln(e^t) = \ln(1 + \sqrt{2}) \quad e^t > 0$$

$$t = \ln(1 + \sqrt{2})$$

$$\text{only}$$

$$\therefore \text{no solution}$$

$$\text{b) } \cos 2\theta = \cos^2 \theta - \sin^2 \theta$$

$$= (1 - \sin^2 \theta) - \sin^2 \theta$$

$$= 1 - 2\sin^2 \theta$$

$$2\sin^2 \theta = 1 - \cos 2\theta$$

$$\sin^2 \theta = \frac{1}{2} (1 - \cos 2\theta)$$

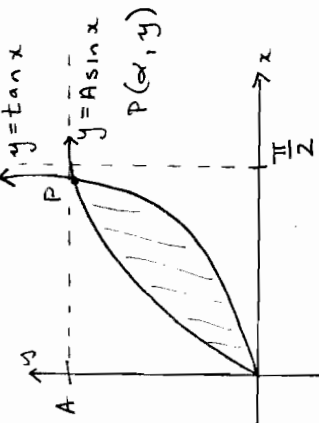
$$\therefore \int \sin^2 3x \, dx$$

$$= \frac{1}{2} \int (1 - \cos 6x) \, dx$$

$$= \frac{1}{2} \left[x - \frac{1}{6} \sin 6x \right] + C$$

$$= \frac{x}{2} - \frac{1}{12} \sin 6x + C$$

c)



ii) P pt of intersection

$$\therefore \tan \alpha = A \sin \alpha$$

$$\frac{\sin \alpha}{\cos \alpha} = A \sin \alpha$$

$$\therefore \frac{1}{\cos \alpha} = A \Rightarrow \cos \alpha = \frac{1}{A}$$

$$\text{iii) } \Delta x = \int_0^\alpha (A \sin x - \tan x) \, dx$$

$$= \int_0^\alpha \left(A \sin x - \frac{\sin x}{\cos x} \right) dx$$

$$= \left[-A \cos x + \ln(\cos x) \right]_0^\alpha$$

$$= (-A \cos \alpha + \ln(\cos \alpha)) - (-A)$$

$$= -A \cdot \frac{1}{A} + \ln\left(\frac{1}{A}\right) + A$$

$$\Delta \text{Area} = -1 + \ln A + A = (A - 1 - \ln A) \text{ unit}^2$$